

Maven Archetype – Advanced Interview Notes

1) What is Maven Archetype?

A Maven Archetype is a reusable project template that generates a complete project structure with configurations, dependencies, and sample code. It ensures standardization and saves setup time.

2) Command to Generate Project

```
mvn archetype:generate
```

3) What Happens Internally?

Step	Explanation
1	User provides groupId, artifactId
2	Maven downloads archetype
3	Folder structure is created
4	pom.xml is generated
5	Dependencies + sample code added

Flow Diagram

```
graph TD
    A[User Input] --> B[Select Archetype]
    B --> C[Download Template]
    C --> D[Generate Structure]
    D --> E[pom.xml + Code Ready]
```

4) Spring MVC Archetype (Traditional Web App)

Used for creating web applications using DispatcherServlet and manual configuration.

Structure

```
project/
  ■■■ src/main/java
  ■■■ src/main/webapp
  ■   ■■■ WEB-INF/
```

```
■ ■ ■ web.xml
■ ■ ■ dispatcher-servlet.xml
■ ■ ■ index.jsp
■ ■ ■ pom.xml
```

pom.xml Example

```
<dependency>
  <groupId>org.springframework</groupId>
  <artifactId>spring-webmvc</artifactId>
</dependency>
```

Key Files

web.xml → Front controller Dispatcher-servlet.xml → Spring config JSP → UI

5) Spring Boot Archetype (Modern Approach)

Creates production-ready application with auto-configuration and embedded server.

Structure

```
project/
  ■ ■ ■ src/main/java
  ■ ■ ■ Application.java
  ■ ■ ■ src/main/resources
  ■ ■ ■ application.properties
  ■ ■ ■ pom.xml
```

pom.xml Example

```
<dependency>
  <groupId>org.springframework.boot</groupId>
  <artifactId>spring-boot-starter-web</artifactId>
</dependency>
```

Key Files

Application.java → Main class application.properties → Config Embedded Tomcat included

6) Difference: Spring MVC vs Spring Boot

Feature	Spring MVC	Spring Boot
Config	Manual XML	Auto
Server	External	Embedded

Setup	Complex	Simple
Speed	Slower setup	Faster

7) Advantages of Archetypes

• Faster project creation • Standard structure • Reduces errors • Easy onboarding

8) Interview Q&A;

Q1: What is Archetype?

Template to generate project structure.

Q2: Can we create custom archetype?

Yes, organizations create their own templates.

Q3: Difference between Archetype and Dependency?

Archetype creates project structure, Dependency adds libraries.

Q4: Why use Spring Boot instead of MVC?

Less configuration, faster development, embedded server.